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EXPERIMENT – 7

Performing Code Injection on Application Process Using Ptrace in Kali Linux

Aim

To demonstrate code injection on a running application process using the ptrace system call in Kali Linux for understanding low-level process manipulation and security vulnerabilities.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Programming Language:** C
- **Debugger / System Tool:** Ptrace
- **Compiler:** GCC
- **System Requirements:**
  - RAM: Minimum 4 GB (8 GB recommended)
  - Disk Space: Minimum 30 GB
  - Processor: Intel/AMD with Virtualization support

Introduction

Code injection is a technique where malicious code is inserted into a running process to alter its execution. ptrace is a Linux system call that allows one process to observe and control another process, commonly used by debuggers. Understanding ptrace-based code injection helps analyze malware behavior, reverse engineering techniques, and system-level security vulnerabilities. Kali Linux provides a secure environment to study such low-level attacks ethically.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Create a Target Program

1. Open Terminal.
2. Create a simple C program:
3. nano target.c
4. Write a basic infinite loop program.
5. Compile the program: 6. gcc target.c -o target 7.

Step 3: Run the Target Application

1. Execute the program:
2. ./target
3. Note the **Process ID (PID)** using:
4. ps aux | grep target **Step 4:**

Create Injector Program

1. Create injector source file:
2. nano injector.c
3. Write ptrace-based code to attach to the target process and modify its memory or registers.
4. Save the file.

Step 5: Compile the Injector Program

1. Compile the injector:
2. gcc injector.c -o injector

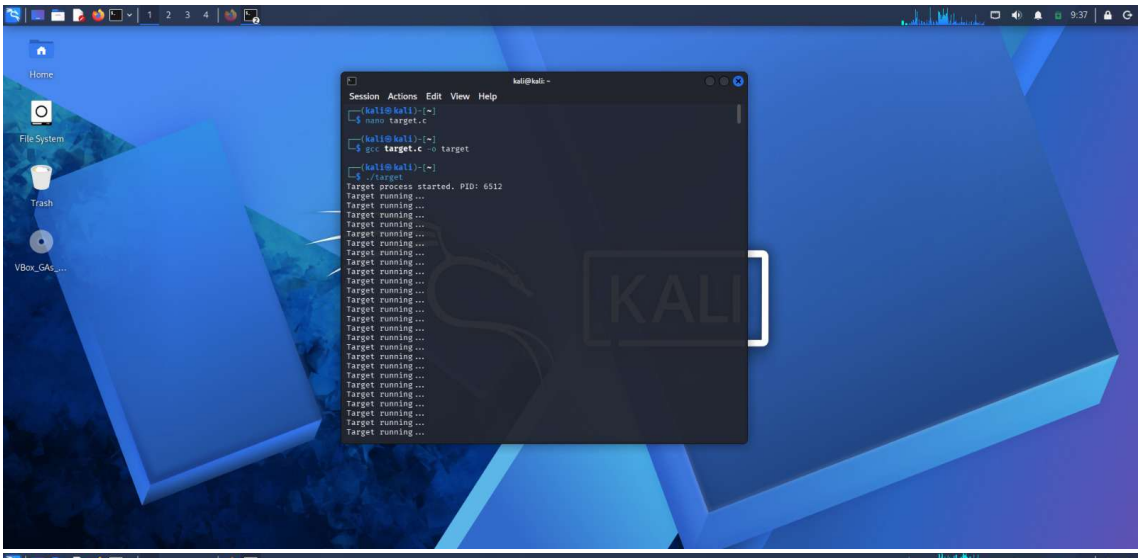
Step 6: Attach to Target Process

1. Attach injector to target process using PID:
2. sudo ./injector <PID>
3. Injector gains control of the target process.

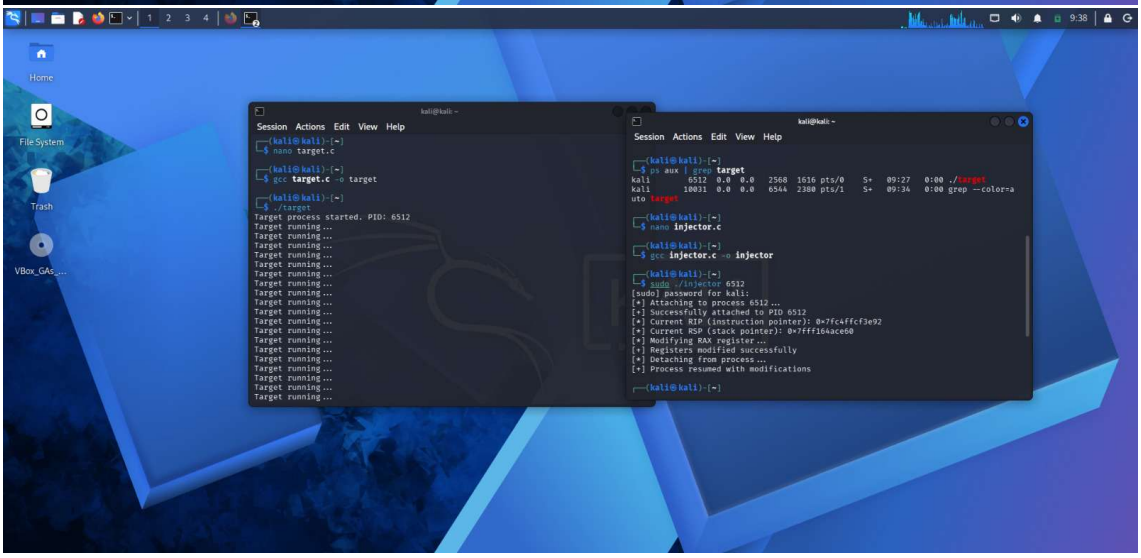
Step 7: Inject and Execute Code

1. Modify target process memory or registers.
2. Resume execution of the target process.
3. Observe injected behavior.

Output


kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ nano target.c
┌──(kali㉿kali)-[~]
└─$ gcc target.c -o target
┌──(kali㉿kali)-[~]
└─$ ./target
Target process started. PID: 6512
Target running ...

kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ nano target.c
┌──(kali㉿kali)-[~]
└─$ gcc target.c -o target
┌──(kali㉿kali)-[~]
└─$ ./target
Target process started. PID: 6512
Target running ...


kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ nano target.c
┌──(kali㉿kali)-[~]
└─$ gcc target.c -o target
┌──(kali㉿kali)-[~]
└─$ ./target
Target process started. PID: 6512
Target running ...
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ ps aux | grep target
kali 6512 0.0 0.0 2568 1616 pts/0 S+ 09:27 0:00 ./target
kali 10031 0.0 0.0 6544 2380 pts/1 S+ 09:34 0:00 grep --color=auto target
┌──(kali㉿kali)-[~]
└─$ nano injector.c
┌──(kali㉿kali)-[~]
└─$ gcc injector.c -o injector
┌──(kali㉿kali)-[~]
└─$ sudo ./injector 6512
[sudo] password for kali:
[*] Attaching to process 6512 ...
[+] Successfully attached to PID 6512
[*] Current RIP (instruction pointer): 0x7fc4ffcf3e92
[*] Current RSP (stack pointer): 0x7fff164ace60
[*] Modifying RAX register ...
[+] Registers modified successfully
[*] Detaching from process ...
[+] Process resumed with modifications
┌──(kali㉿kali)-[~]

Result

Code injection using the ptrace system call was successfully demonstrated in Kali Linux. The experiment illustrates how process memory and execution flow can be manipulated, emphasizing the importance of secure process isolation.